

Cyclic AMP waves during aggregation of *Dictyostelium* amoebae

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Summary

During the aggregation phase of their life cycle, *Dictyostelium discoideum* amoebae communicate with each other by traveling waves of cyclic AMP. These waves are generated by an interplay between random diffusion of cyclic AMP in the extracellular milieu and the signal-reception/signal-relaying capabilities of individual amoebae. Kinetic properties of the enzymes, transport proteins and cell-surface receptor proteins involved in the cyclic AMP signaling system have been painstakingly worked out over the past fifteen years in many laboratories. Recently Martiel & Goldbeter (1987) incorporated this biochemical information into a unified mathematical model of communication among *Dictyostelium*

amoebae. Numerical simulations of the mathematical model, carried out by Tyson *et al.* (1989), agree in quantitative detail with experimental observations of cyclic AMP traveling waves in *Dictyostelium* cultures. Such mathematical modeling and numerical experimentation provide a necessary link between detailed studies of the molecular control mechanism and experimental observations of the intact developmental system.

Key words: *Dictyostelium discoideum*, cyclic AMP, target patterns, spiral waves, aggregation, mathematical models, numerical experiments.

Biological background

When amoebae of the myxomycete *Dictyostelium discoideum* are left to starve on an agar surface, they begin signaling to each other with a chemical messenger, cyclic adenosine 3',5'-monophosphate (cAMP) (Bonner, 1969). Individual cells receive the signal by binding extracellular cAMP to a membrane receptor (Newell, 1986), and this binding stimulates the synthesis of cAMP from ATP by adenylate cyclase within the cell. Newly synthesized cAMP is transported to the extracellular medium and in this fashion the chemical signal is amplified. Amplification of the signal is limited by the fact that, on prolonged exposure to cAMP, the membrane receptor becomes desensitized (Devreotes & Sherring, 1985), i.e. it no longer stimulates adenylate cyclase activity. In the absence of cAMP synthesis, the concentration of cAMP decreases by the action of phosphodiesterase, which hydrolyzes cAMP to 5'-AMP.

In a field of signaling amoebae spread over an agar surface, pulses of extracellular cAMP travel across the field in the form of either expanding concentric circular waves or rotating spiral waves. As the waves pass periodically through the field of independent amoebal cells, they stimulate a chemotactic movement of the amoebae towards the center of the pattern (the origin of the circles or the pivot point of the spiral). Eventually

all the amoebae within the domain of a single pattern aggregate at the center to form a multicellular slug, which goes on to form a fruiting body. Fig. 1 shows typical spiral and target patterns in a field of aggregating amoebae. The aggregation phase of the life cycle of *Dictyostelium discoideum* is often taken as a paradigm for developmental biology (Bonner, 1961; Loomis, 1975, 1982). In it we see the development of a spatio-temporally periodic chemical prepattern that induces a pattern of morphogenetic movements culminating in the formation of an organized multicellular tissue. A thorough understanding of this simple developmental process at the molecular level is possible because of our detailed knowledge of the molecular mechanism of the cAMP relay response.

To achieve such understanding it is necessary to study experimentally the kinetics of cAMP-receptor binding, of adenylate cyclase activity, and of phosphodiesterase activity. However, studies of the isolated components are not sufficient to determine exactly how the full system functions. To construct a scenario of the coordinated activity of the complete mechanism, we must develop a mathematical model of the cAMP signal-relaying system, determine the model's behavior by analytical and numerical methods, and compare the behavior of the model in quantitative detail with experimental measurements on the intact cellular system.

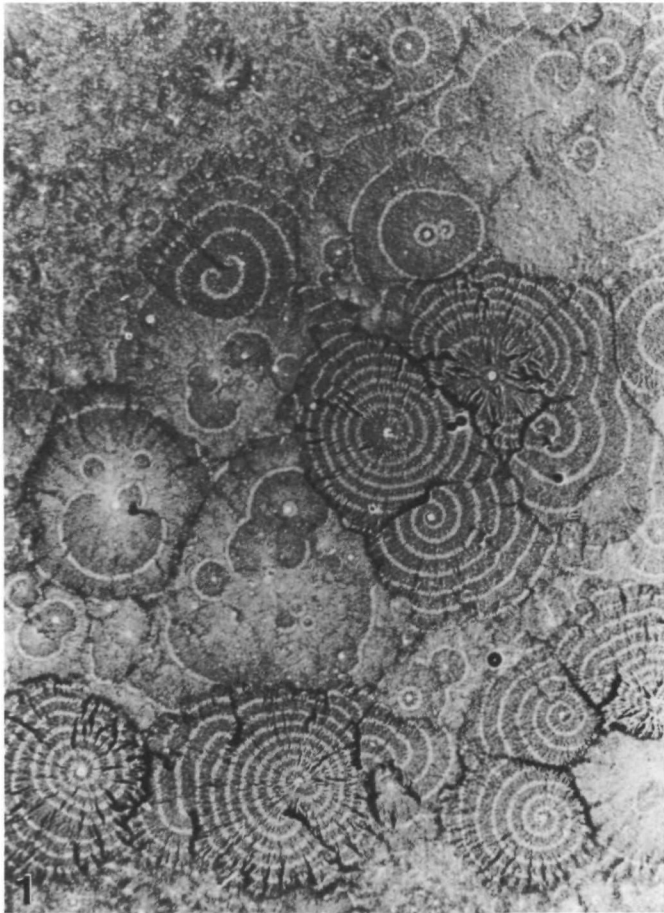


Fig. 1. Target and spiral patterns in *Dictyostelium discoideum* (from Newell, 1983, courtesy of P. C. Newell).

Early models

Many different models of cell-to-cell communication in *Dictyostelium* have been proposed over the years. The earliest models were necessarily a little crude because essential biochemical information about the signaling system was lacking. For instance, before the kinetic properties of the membrane receptor were characterized, Goldbeter & Segel (1977, 1980) suggested a model in which intracellular ATP, intracellular cAMP, and extracellular cAMP were the essential dynamical variables. By assuming positive feedback of extracellular cAMP on the activity of adenylate cyclase, Goldbeter and Segel were able to demonstrate oscillations and signal relaying in the model that were remarkably similar to experimental observations in well-stirred cell-suspension cultures. Unfortunately, however, the Goldbeter–Segel model predicts noticeable fluctuations in intracellular ATP concentrations in the oscillatory mode, and such fluctuations have never been observed. More seriously, the Goldbeter–Segel model was unable to account for adaptation of the cAMP response to repeated stimulation of *Dictyostelium* cells by external application of cAMP (Dinauer *et al.* 1980a,b). An alternative model, suggested by Rapp & Berridge (1977), attributed cAMP oscillations to inter-

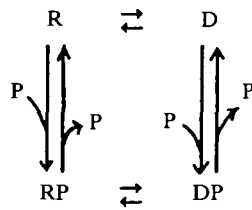
actions between internal calcium and cAMP. Recent elaboration of this idea by Rapp *et al.* (1985) gives an impressive account of signal-relay adaptation by *Dictyostelium* cells in suspension culture, but to date there is no direct evidence in *Dictyostelium* for the postulated interactions between calcium and cAMP.

Modeling cAMP wave propagation in agar-surface cultures introduces spatial dependencies and is much more complicated than modeling oscillations and signal relaying in cell-suspension cultures because as well as the signal-relaying system we now have to account for random diffusion of cAMP in the extracellular milieu, random motion of the amoebae on the agar surface, and chemotactic motion of the amoebae. The earliest model (Cohen & Robertson, 1971a,b) was based on simple rules for synthesis, release, degradation and diffusion of cAMP, and movements of amoebae. This idea was later developed into remarkable computer simulations of the aggregation process in one and two spatial dimensions (Parnas & Segel, 1977, 1978; MacKay, 1978). An alternative to such rule-based computer simulations is to describe the aggregation field by a set of reaction–diffusion equations which determine the spatial and temporal evolution of amoebal cell density and extracellular cAMP concentration. In a pioneering paper, Keller & Segel (1970) showed, with a simple but effective model of cAMP turnover, that the initiation of slime mold aggregation can be viewed as an instability in the homogeneous solution of such partial differential equations. Later, Hagan & Cohen (1981) showed that, through a sequence of stability changes, a reaction–diffusion model of *Dictyostelium* development can exhibit an impressive sequence of morphogenetic behaviour such as pulse relaying, spiral waves, target patterns, cell streaming and sorting, slug locomotion, and tissue buckling. Unfortunately, this valuable paper is not easily accessible to most developmental biologists because of the technical complexity of the mathematical analysis.

The receptor–cAMP interaction

Although early models can be criticized for making unconfirmed assumptions about the biochemistry of cAMP metabolism, they uncovered many qualitative properties of the cAMP signaling system that remain valid even in light of our present knowledge of *Dictyostelium* biochemistry. However, to obtain a model that is quantitatively accurate and biochemically convincing, we must have precise, quantitative biochemical information about the control system, especially about the membrane receptor for cAMP. Such information has only recently been obtained by the studies of Van Haastert & DeWit (1984) and Devreotes & Sherring

(1985). From these studies we are presented a picture of the receptor–cAMP interaction as a ‘receptor box’:



The receptor exists in four forms: R, unbound and active; D, unbound and inactive; RP, bound and active; DP, bound and inactive. Based on the receptor-box picture of signal reception in *Dictyostelium*, Martiel & Goldbeter (1987) developed a complete model of cAMP signaling by adding several reactions summarizing the synthesis of cAMP by adenylate cyclase, the transport of cAMP across the plasma membrane, and the degradation of cAMP by phosphodiesterase. By classical methods of biochemical kinetics, Martiel & Goldbeter (1987) derived a set of rate equations describing the dynamical interactions of intracellular cAMP, extracellular cAMP and the membrane receptor. Using experimental measurements of the rate constants and binding constants for all the known steps of the reaction mechanism and choosing reasonable values for the few unknown parameters in the model, they simulated the behavior of the system by numerical solution of the reaction kinetic equations. For certain parameter values, their model exhibits autonomous oscillations of cAMP which agree in quantitative detail with experimental observations of the period, amplitude, and waveform of cAMP oscillations in well-stirred suspensions of *Dictyostelium* cells. For slightly different parameter values, the model predicts that cells will respond to an external pulse of cAMP by one-time amplification of the cAMP pulse. Model calculations agree in quantitative detail with experimental observations of the relay response of *Dictyostelium* cells in suspension, of adaptation to constant stimulation, and of the response of cells to periodic pulsatile and stepwise stimulation (Martiel & Goldbeter, 1987). Similar models of oscillations, relay and adaptation, based on receptor modification, have been presented by Segel *et al.* (1986) and by Barchilon & Segel (1988).

cAMP waves

To model cAMP waves in agar-surface cultures of *Dictyostelium* cells, Tyson *et al.* (1989) investigated a set of reaction–diffusion equations consisting of the Martiel–Goldbeter reaction kinetics for cAMP signaling combined with diffusion of cAMP through the extracellular milieu. Their model describes spatial and temporal variations in the extracellular cAMP concentration (γ), the intracellular cAMP concentration (β), and the fraction of membrane receptor in the active form (ρ). The model consists of an equation for each of

these three variables, which we give first in words and then in mathematical form underneath:

$$\begin{array}{lcl} \text{Rate of change} & = & \text{secretion} \\ \text{of extracellular} & = & \text{by the} \\ \text{cAMP} & = & \text{cells} \\ & - & \text{extra-} \\ & & \text{cellular} \\ & & \text{hydrolysis} \\ & + & \text{random} \\ & & \text{diffusion} \end{array} \quad \frac{\partial \gamma}{\partial t} = (k_i/h)\beta - k_e\gamma + D\nabla^2\gamma \quad (1)$$

$$\begin{array}{lcl} \text{Rate of change} & = & \text{synthesis} \\ \text{of intracellular} & = & \text{in the} \\ \text{cAMP} & = & \text{cells} \\ & - & \text{secretion} \\ & & \text{by the} \\ & & \text{cells} \\ & - & \text{intra-} \\ & & \text{cellular} \\ & & \text{hydrolysis} \end{array} \quad \frac{\partial \beta}{\partial t} = q\Phi(\rho, \gamma) - k_i\beta - k_e\beta \quad (2)$$

$$\begin{array}{lcl} \text{Rate of change} & = & \\ \text{of active form} & = & \\ \text{of receptor} & = & \\ & - & \text{desensitiz-} \\ & & \text{ation of} \\ & & \text{receptor} \\ & + & \text{resensitiz-} \\ & & \text{ation of} \\ & & \text{receptor} \end{array} \quad \frac{\partial \rho}{\partial t} = -f_1(\gamma)\rho + f_2(\gamma)(1-\rho) \quad (3)$$

In these equations,

$$\rho = \frac{[R] + [RP]}{[R] + [RP] + [D] + [DP]}$$

where $[R]$, $[RP]$, $[D]$ and $[DP]$ are concentrations of the various forms involved in the receptor box described in the previous section. D is the diffusion coefficient of extracellular cAMP, and ∇^2 is the diffusion operator (second spatial derivatives). The constants k_i , k_e , q and k_i are rate constants associated with the metabolism of cAMP, and h is the ratio of extracellular volume to intracellular volume. The rate functions $f_1(\gamma)$ and $f_2(\gamma)$ describe the kinetics of the receptor box, and $\Phi(\rho, \gamma)$ describes the activation of adenylate cyclase by bound and active receptor (RP). The forms of these functions are given by Martiel & Goldbeter (1987) as

$$f_1(\gamma) = \frac{k_1 + k_2\gamma}{1 + \gamma}, \quad f_2(\gamma) = \frac{k_{-1} + k_{-2}c\gamma}{1 + c\gamma},$$

$$\Phi(\rho, \gamma) = \frac{\lambda_1 + Y^2}{\lambda_2 + Y^2}, \quad Y = \frac{\rho\gamma}{1 + \gamma},$$

where the k 's, λ 's and c are constant parameters.

Tyson *et al.* (1989) solved the system of equations (1)–(3) numerically with specific values for the various constants gleaned as far as possible from experimental data (Martiel & Goldbeter, 1987). They calculated two sorts of cAMP waves. The first sort are planar uncurved waves propagating periodically through a field of cells (Fig. 2). In this case, the speed of propagation of the cAMP waves depends on the temporal periodicity of the wave train, as indicated in Fig. 3. For periods down to approx. 6 min the wave speed is constant at 0.28 mm min^{-1} . At shorter periods the wave speed decreases abruptly because there is not enough time between waves for the membrane receptor to recover full sensitivity. Below the minimum period (4.5 min) no wave trains are possible because the membrane receptor cannot recover fast enough between such high-frequency pulses.

The calculated relation between wave speed and

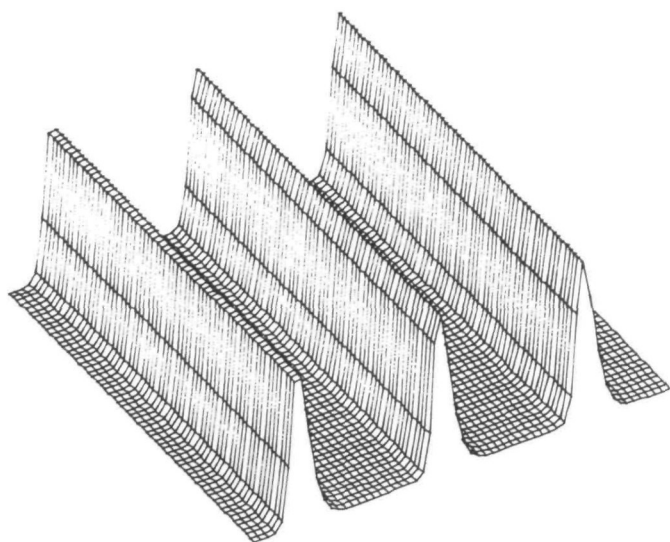


Fig. 2. Plane waves in two dimensions. The concentration of extracellular cAMP is illustrated in this snapshot of a wave train. The waves are moving from right to left at speed 0.28 mm min^{-1} . The wavelength of this pattern is 2.3 mm ; the period is 8 min .

period illustrated in Fig. 3 is germane to expanding concentric circular waves of cAMP. Sufficiently far from the source of such 'target' patterns, the traveling waves of cAMP have negligible curvature and resemble the uncurved periodic waves of Fig. 2. Therefore, the temporal period and wave speed of a target pattern must satisfy the relation in Fig. 3. Alcantara & Monk (1974) measured the velocity of waves in target patterns as a function of period and found the wave velocity to be roughly constant ($0.256 \text{ mm min}^{-1}$) for temporal periods between 4 and 10 min. Their measurements are plotted in Fig. 3 along with the theoretical curve for the MG model derived by Tyson *et al.* (1989): the agreement between theory and experiment is gratifying.

In a second set of calculations, Tyson *et al.* (1989)

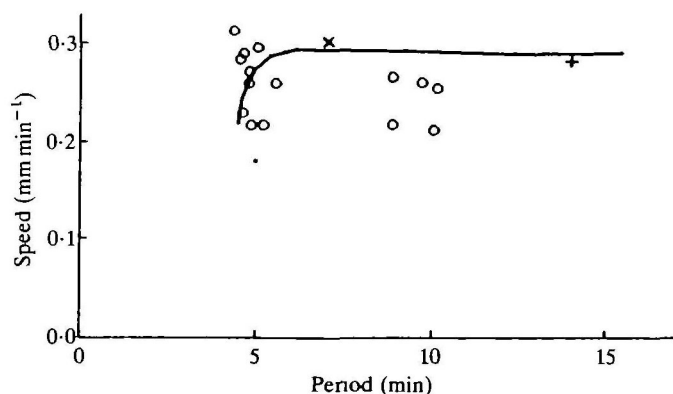


Fig. 3. Relation between wave speed and period for plane waves. Solid line: as period predicted by MG model (Tyson *et al.* 1989). Open circles: wave speed as measured by Alcantara & Monk (1974). (+): spiral wave calculated for MG model (Tyson *et al.* 1989). (X): spiral wave observed by Tomchik & Devreotes (1981).

looked for rotating spiral wave solutions to the model equations (1)–(3): a snapshot of a typical spiral wave is illustrated in Fig. 4. In contrast to target patterns, which can occur at any temporal period greater than 4.5 min , there is only one spiral wave solution, which rotates with a period of 14 min and has a wave speed of 0.28 mm min^{-1} . These computed values compare favorably with the cAMP spiral wave observed by Tomchik & Devreotes (1981), Fig. 5, which had a rotation period of 7 min and a wavespeed of 0.3 mm min^{-1} . Furthermore, the amplitude of the computed spiral wave ($[\text{cAMP}]_{\text{max}} = 10^{-6} \text{ M}$, $[\text{cAMP}]_{\text{min}} = 3 \times 10^{-9} \text{ M}$) agrees well with the measured amplitude ($[\text{cAMP}]_{\text{max}} = 10^{-6} \text{ M}$, $[\text{cAMP}]_{\text{min}} \leq 5 \times 10^{-8} \text{ M}$; Devreotes *et al.* 1983).

The model described by equations (1)–(3) neglects any contribution from the motion of amoebae in response to the traveling waves of cAMP, so the model is valid only in the initial stages of aggregation when there is still a uniform distribution of amoebae on the agar surface. To describe later stages of the process, when the amoebal distribution becomes nonuniform, will require a more complicated model including the effects of chemotaxis and cell adhesion (Keller & Segel, 1970, 1971; Parnas & Segel, 1977, 1978).

Excitable media

Cell–cell signaling by cAMP in *Dictyostelium* is one instance of a more general principle of spatial organization by traveling waves of 'excitation' in signal-relaying systems (Durstion, 1973). The propagation of action potentials along nerve axons is the most familiar example, but other situations bear more resemblance to the *Dictyostelium* case. Waves of neuromuscular activity spread through heart muscle and, under certain

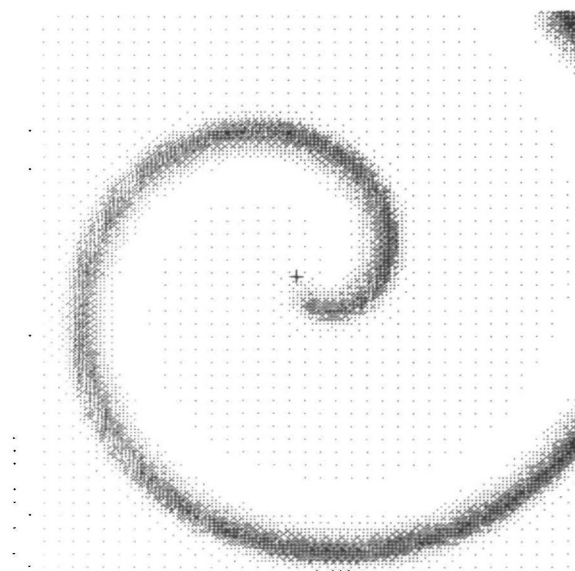


Fig. 4. Spiral wave of extracellular cAMP calculated by Tyson *et al.* (1989).

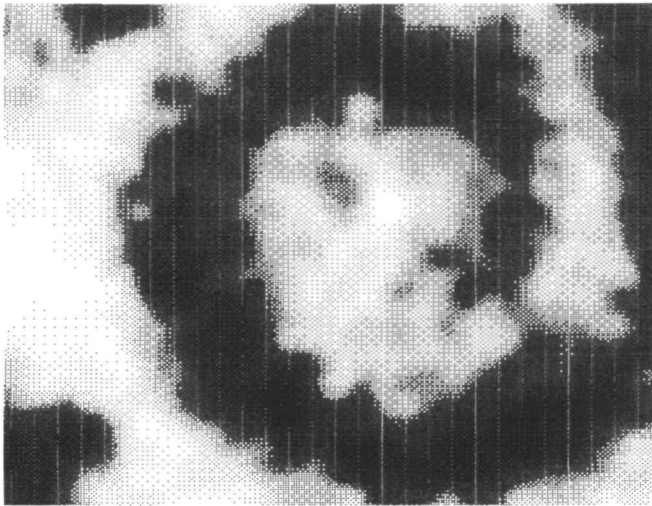


Fig. 5. Spiral wave of extracellular cAMP measured by Tomchik & Devreotes (1981).

conditions, these waves of muscular contraction may take the form of expanding target patterns or rotating spiral waves (Winfree, 1987). Rotating spiral waves are also observed on the cerebral cortex (Petsche *et al.* 1974) and on the retina of the eye (Gorelova & Bures, 1983). Recurrent waves of infection spread spatially through susceptible populations (Carey *et al.* 1978; Murray *et al.* 1986). The Belousov-Zhabotinski reaction is a thoroughly studied chemical system that propagates waves of oxidation, exhibiting target patterns and spiral waves in thin layers and complex three-dimensional scroll-shaped waves in deep solutions (Winfree, 1987). The most spectacular example of long-range spatial organization are the waves of star formation in spiral galaxies (Schulman & Seiden, 1986). All of these examples share a common feature of signal-relaying capacity – be it cAMP, transmembrane ionic currents, infectious microbes, chemical oxidation, or stellar nucleation. In all cases the amplification and spatial spreading of the signal can be described by a system of ‘reaction-diffusion’ equations similar to the ones considered in this paper. From analytical and numerical studies of generic models, a general theory of spatial organization in excitable systems is emerging (Tyson & Keener, 1988; Zykov, 1988). Thus, the detailed description of cAMP waves during aggregation of *Dictyostelium* amoebae is important not only for the insight it provides into developmental processes but also for the challenge it presents to our general understanding of spatial organization in excitable media.

Conclusion

The model of Martiel & Goldbeter (1987), based on the ‘receptor box’ kinetics worked out experimentally by Devreotes & Sherring (1985), is remarkably successful in accounting for many features of the cAMP signaling

system in *Dictyostelium* cells. The good quantitative agreement between theory and experiment on cAMP waves during aggregation of *Dictyostelium* amoebae is very encouraging and provides reasonable cause for optimism in modeling the complete developmental process in this organism.

To what extent is this success a paradigm for developmental biology? *Dictyostelium* aggregation is in many respects a special case in that it is a simple process organized by traveling waves and a process for which we have a good description of the kinetics of the biochemical reactions taking place. The temptation to apply the model to other developmental processes must be treated with extreme caution. For example, many steps in embryonic development rely on the formation of spatial patterns, including spirals, but for most of these patterns there is no evidence of wave propagation as exhibited by *Dictyostelium*. Cartilage patterning in the vertebrate limb is one widely studied example. Quite different models are probably required to generate steady-state patterns, as opposed to traveling wave patterns, in development (see, for example, Murray, 1988). In each case of developmental pattern formation, the details of the actual mechanism involved are generally quite different. Though the mechanistic details of *Dictyostelium* aggregation are probably not generalizable, the method of establishing a unified description is. To explain a complex developmental process, unfolding in space and time, it is necessary but not sufficient to study the individual pieces of the molecular machinery. Eventually these pieces must be put together into a mathematical model, and the model must be studied by analytical and numerical methods to demonstrate that the mechanism really can account for the developmental process in quantitative detail.

Mathematical modeling and numerical experimentation, of the sort we have reviewed here, are becoming increasingly important in developmental biology. Their

ultimate success, however, will depend crucially on close interaction between biologists and theoreticians.

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